OFTEn Framework

Participatory activities for purposeful, transparent, and people-centric dataset documentation

OFTEn Framework

What is Knowledge Acquisition?

KNOWLEDGE ACQUISITION (KA)

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is the process of extracting, structuring and organizing knowledge from one source, usually human experts, so it can be used in software such as an Expert Systems (ES).

OFTEn is a knowledge acquisition framework designed to elicit information about any dataset, that is important for transparency in reporting.

OFTEn can be used on any dataset to identify and evaluate content that’s important to your agents.

OFTEn represents general stages in a dataset's lifecycle

ORIGINS	FACTUALS	TRANSFORMATIONS	EXPERIENCE	n = 1
Early stages of a dataset's lifecycle when decisions to create a dataset are made	Actual data collection processes and raw outputs	Raw data is transformed into a usable form through operations like filtering, validating, parsing, formatting, and cleaning	Dataset is tested, benchmarked, or deployed in practice (experimental, production, or research)	Actual samples from the dataset - or vignettes - representing normal datapoints and outliers

ORIGINS

Various planning activities such as defining requirements, design decisions, collection or sourcing methods, and deciding policies which dictate final outcome.

Examples of Questions	Resulting Themes	Example
Who collected it? **Why** was it collected? **What** is it about? **How, where & when** was it collected? **When** was it updated? **Who** maintains it? ...	Authorship Motivations Intended applications Licenses Versions Sources Collection methods Errata Accountable parties	Credit Card Fraud Detection, Kaggle **Who** created this dataset? **Publisher:** Machine Learning Group, ULC **Publisher Type:** Academic - University Industry - Financial

FACTUALS

	Examples of Questions	Resulting Themes	Example
Statistical and other factual attributes that describe the dataset, deviations from the original plan, and any pre-wrangling analysis.	**How** many instances and features? **What** are the labels in this dataset? **Who** labelled it? **What** is the structure and format? **How** should an example be read or interpreted? **Where** is the dataset hosted? ...	Number of instances Number of features Number of labels Source of labels Source of data Breakdown of subgroups Shape of features Description of features Missing or duplicates Inclusion criterion	Credit Card Fraud Detection, Kaggle **Who** does this dataset represent? **Represented Population:** September 2013 European Cardholders **Geographic Distribution:** Europe-only

TRANSFORMATIONS

	Examples of Questions	Resulting Themes	Example
Summaries of labeling, annotation, or validation tasks. Inter-rater adjudication processes. Feature engineering and modifications made to handle privacy, security, or PII.	**How** was this dataset cleaned or verified? **How** was this data annotated, validated, or rated? **Who** did this? **Which** features correlate with each other? **What** are the known sensitive variables? **What** metrics were used to define dataset quality? ...	Rating or annotation Filtering Processing Validation Statistical properties Synthetic features Handling PII Sensitive variables Impact on fairness Skews or biases	Credit Card Fraud Detection, Kaggle **How** was identity handled? **Privacy:** Due to confidentiality issues, background information is not provided. **Feature Description:** All features transformed with PCA; with the exception of “Time” and “Amount.”

EXPERIENCE

	Examples of Questions	Resulting Themes	Example
Using the data for specific tasks, undergoing access training, making modifications to suit the task, acquiring results and comparing to other similar datasets, and noting any expected/ unexpected behaviors.	**Who** has used the dataset? **What** kinds of tasks has it been used for? **What** were the methods, results, or errors? **How** should it be used and not be used? **How** many times has it been used? ...	Intended performance Alternate applications Unexpected performance Caveats Insights Experiences Stories Use & use case evaluation	Credit Card Fraud Detection, Kaggle **Who** has used this dataset in the past? **Known use cases - industry application:** Unknown **Known use cases - research and academia:** Widely used in academia for sample problems to train fraud detection classifiers

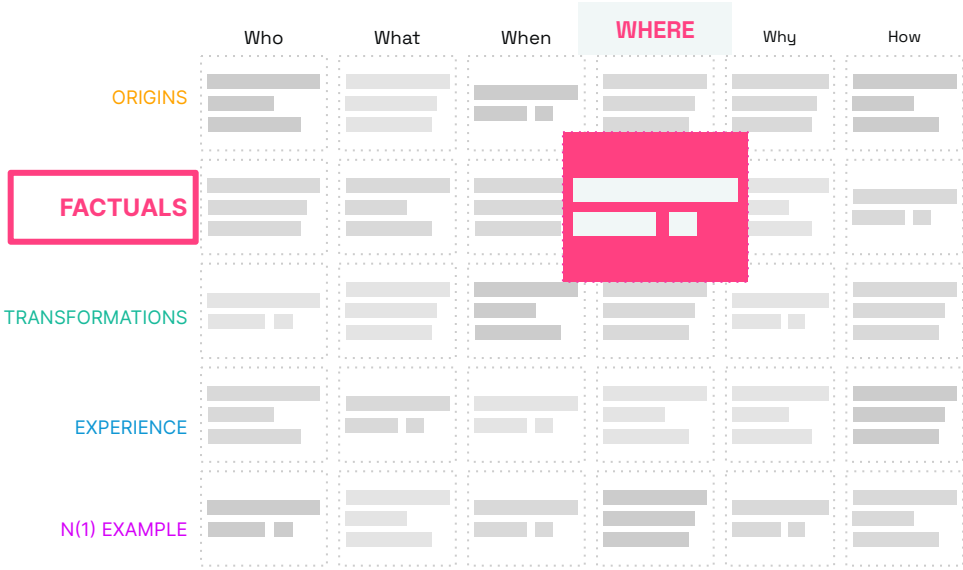
n = 1 (Samples)

	Examples of Questions	Resulting Themes	Example
In-and-out-of distribution datapoints, demonstrates noteworthy data points with specific attributes, and where applicable, model outcomes on them.	**What** does an actual example in this dataset look like? **What** does a prototypical example look like? **Why** is it typical or atypical? **What** is the expected result of a task on this data point?	Examples or links to typical examples and outliers Examples that yield false positives or false negatives Examples that demonstrate handling of null or zero feature values	Credit Card Fraud Detection, Kaggle **Time:** 0 **V1:** -1.3598071336738 **V3:** -0.0727811733098497 **V4:** 2.53634673796914 **V5:** 1.37815522427443 **...** **Amount:** 149.62 **Class:** 0

	Who	What	When	Where	Why	How
ORIGINS	Who is the publisher of the dataset?	What challenges should the dataset have addressed? What are the incentives for data professions, like labellers, practitioners, researchers?	When was this dataset created? Launched?	Where did the funding come from?	Why was this dataset created? What was the process prior?	How were the methods decided? How many parties were involved?
FACTUALS	Who is the data about? Who is responsible for the data? Who are the labellers?	What are the subgroups in the data that can affect outcomes?	What time period does the data represent? When does the data perform abnormally?	Where can the dataset be accessed? Where was the data collected or model created?	Why were the reported metrics chosen?	How many unique labels exist in the dataset? How were these generated?
TRANSFORMATIONS	How was PII handled in this dataset?	What methods were used to clean or verify this dataset? What categories are labelled in all?	When should features be engineered and how? Do these need to be updated?	Do location features correlate with any other sensitive features? Where is the data safe to use?	Why were the chosen transformations applied to the dataset?	How are biases handled in the data?
EXPERIENCE	Can outcomes from this dataset be used to identify individuals? Who can use this dataset? Are there any trainings required?	In practice, what tasks has data been used for? What were the methods, results, or errors discovered when the dataset was used?	Under what circumstances or use cases, and when would this dataset not be effective?	Where in the world is this dataset accessible? Where has it been used?	Why is the expected representation of the dataset different from the observed representation?	In practice, how many times has this dataset been used? How expensive is the data in various parts of the world?
n = 1 (Samples)	Is the datapoint typical or atypical? How do models behave here?	What is the size of the datapoint? What's the consent, redaction, and withdrawal process to intervene in a datapoint?	When does the outcome on a datapoint change? Show examples through counterfactuals?	What factors are baked into the datapoint? What are the risks involved if things go wrong with predictions?	Why is this image datapoint cropped a certain way? Why are certain categories not populated in this datapoint?	How does this datapoint relate to a real-world input? How does the outcome relate to a real-world output?

Try it yourself

Frame a question using OFTEn.



Rows represent life cycle stages - O,F,T,E, and n.
Columns represent prompts for questions.
Frame a question about “Factuals,” starting with “Where.”

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